**EXP NO:8 Conduct Z test for the Data given using Python**

**AIM:**
**To conduct a Z test for the data given using Python.**

**PROGRAM:**

**python**

```python
import numpy as np
from scipy.stats import norm
from statsmodels.stats.weightstats import ztest

# Marks of 40 students
np.random.seed(1)
marks = np.random.normal(loc=72, scale=10, size=40).round()
print(marks)

# Hypotheses
# H0: Population mean = 70
# H1: Population mean is not equal to 70
mu0 = 70        # claimed population mean
sigma = 10      # known population standard deviation
alpha = 0.05    # level of significance

# Sample details
n = len(marks)
sample_mean = marks.mean()
print("Sample Size:", n)
print("Sample Mean:", sample_mean)

# Z statistic
z = (sample_mean - mu0) / (sigma / np.sqrt(n))
print("Z Statistic:", z)

# Two-tailed p-value
p_value = 2 * (1 - norm.cdf(abs(z)))
print("P Value:", p_value)

# Critical values
z_critical = norm.ppf(1 - alpha / 2)
print("Critical Value: +/-", z_critical)

# Decision
if p_value < alpha:
    print("Reject the null hypothesis")
else:
    print("Fail to reject the null hypothesis")

# Verification using the statsmodels library
z_stat, p_val = ztest(marks, value=mu0)
print("Z Statistic (statsmodels):", z_stat)
print("P Value (statsmodels):", p_val)
```

**OUTPUT:**

```
[88. 66. 67. 61. 81. 49. 89. 64. 75. 70. 87. 51. 69. 68. 83. 61. 70. 63.
 72. 78. 61. 83. 81. 77. 81. 65. 71. 63. 69. 77. 65. 68. 65. 64. 65. 72.
 61. 74. 89. 79.]
Sample Size: 40
Sample Mean: 71.05
Z Statistic: 0.6640783086353579
P Value: 0.506640192469326
Critical Value: +/- 1.959963984540054
Fail to reject the null hypothesis
Z Statistic (statsmodels): 0.6803299484617115
P Value (statsmodels): 0.4962955656604181
```

**RESULT:**
**Thus the Z test for the given data was conducted using Python and the null hypothesis was tested at the 5% level of significance.**